

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart, carry the height station by station, return — that gap is f.

Within tolerance, every station on the line is accepted;

over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:

what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.

Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²

Overall design area

9,443 m

Spine length

8

Tiered benchmarks

13.8 km/km²

Network density, measured

17

Oversized blocks, not parks

19.7%

Green corridor, share of design area

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

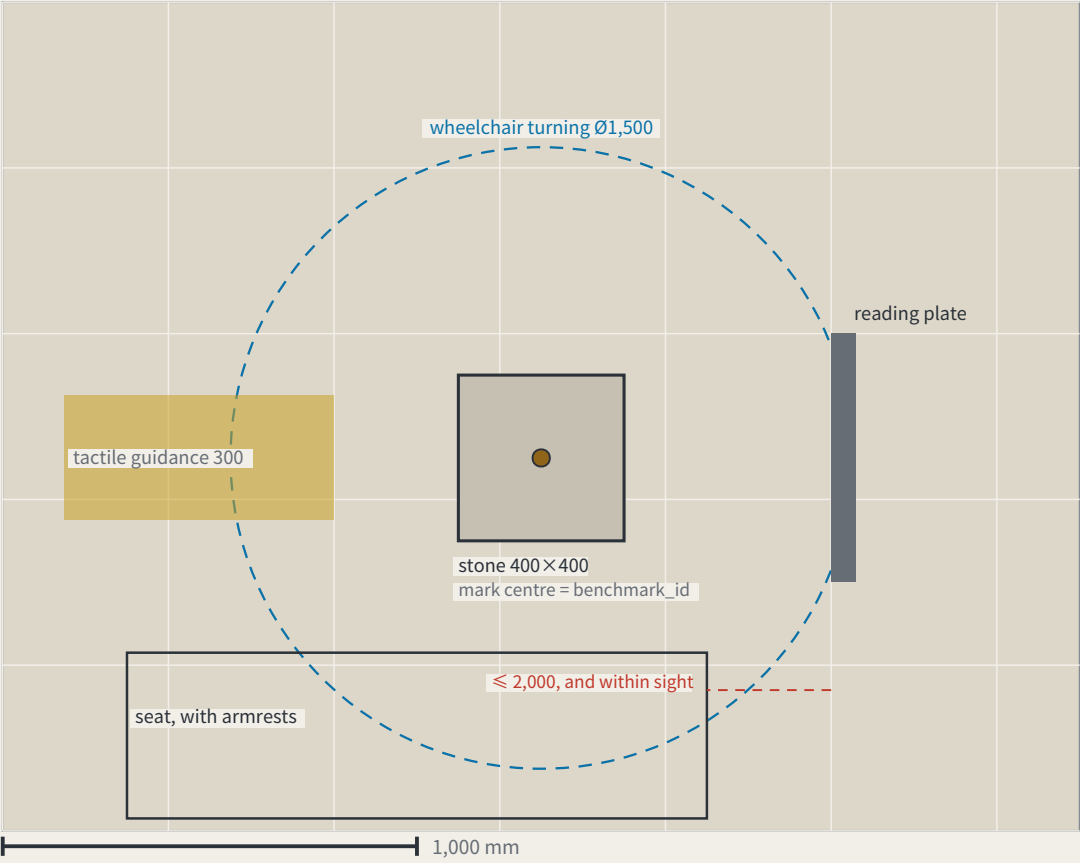
The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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a type drawing; it carries no orientation

I. Plan

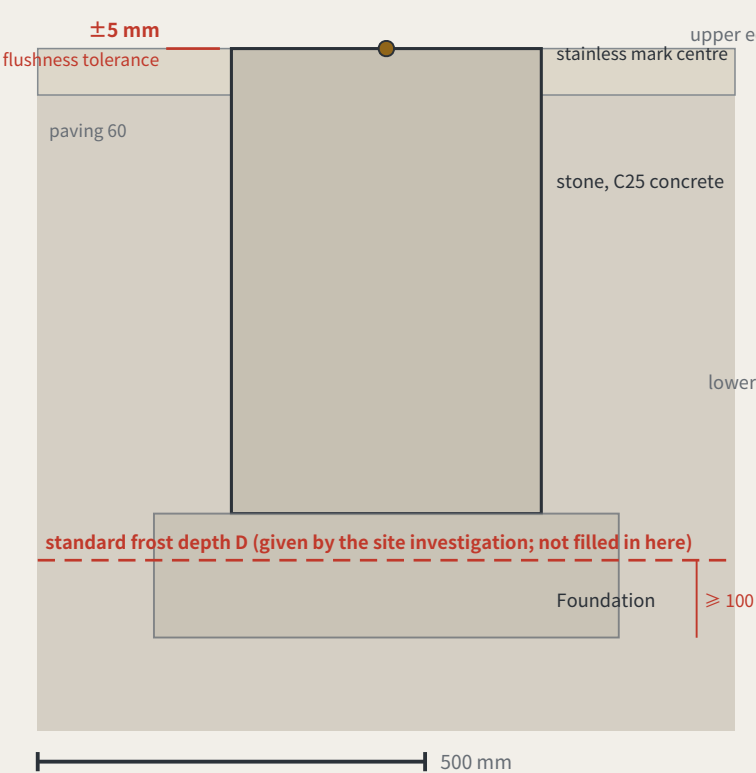


Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

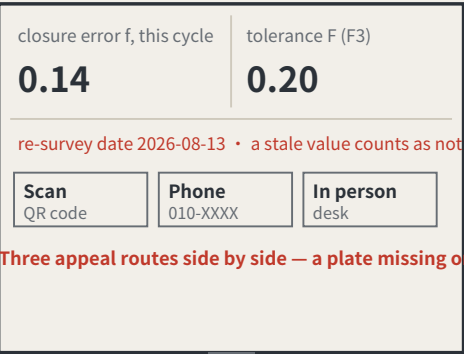
Item	Design intent	Why it is that number	As measured
stone	400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	≤ ±5 mm	KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion	
Depth to underside of foundation	≥ the local standard frost depth D + 100 mm	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	600 × 450 mm, raked 15°	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	900 mm / 1,350 mm	the band a seated and a standing reader share; P5’ s continuous step-free route should not end at a plate they cannot read	
Appeal panel	QR code, phone number and in-person address, all three side by side	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	300 mm wide, stopping 300 mm short of the stone	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	Ø1,500 mm, headroom ≥ 2,100 mm	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	seat 450 mm high, with armrests, within 2,000 mm of the plate	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario_cards.json

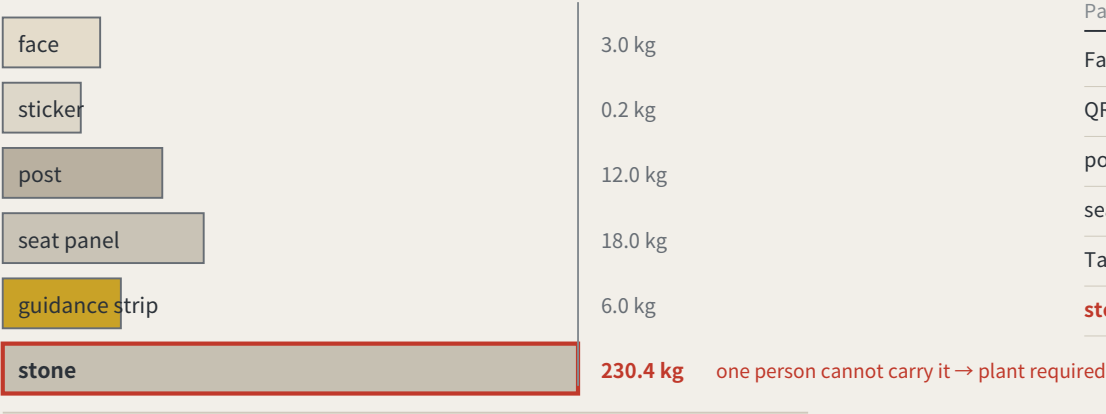


Three appeal routes side by side — a plate missing one fails inspection

The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

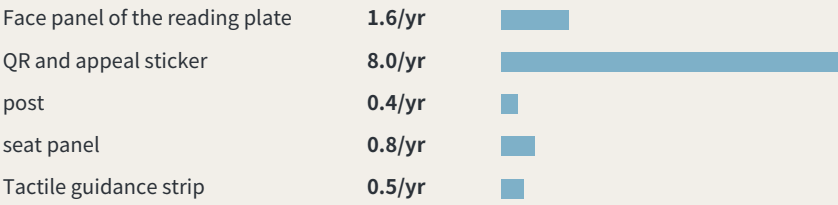
a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved



IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
stone	impact, excavation, settlement	never replaced	230 kg	plant required	yes

III. After it fails: found → held → replaced → lost

Found	any re-survey that finds the face stale or missing records it in that cycle the plate’ s own rule — a stale value counts as not posted — is the detector
Held	the point is marked returned in datum red, shown at every third-order point in the area bad news belongs where it happened (errata E50)
Replaced	a standard part is swapped without touching the stone or its mark centre one person, one visit; heaviest part 18 kg
Stone lost	the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused reusing an id would quietly join two different points into one history

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Closure error in the site's own data

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
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Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

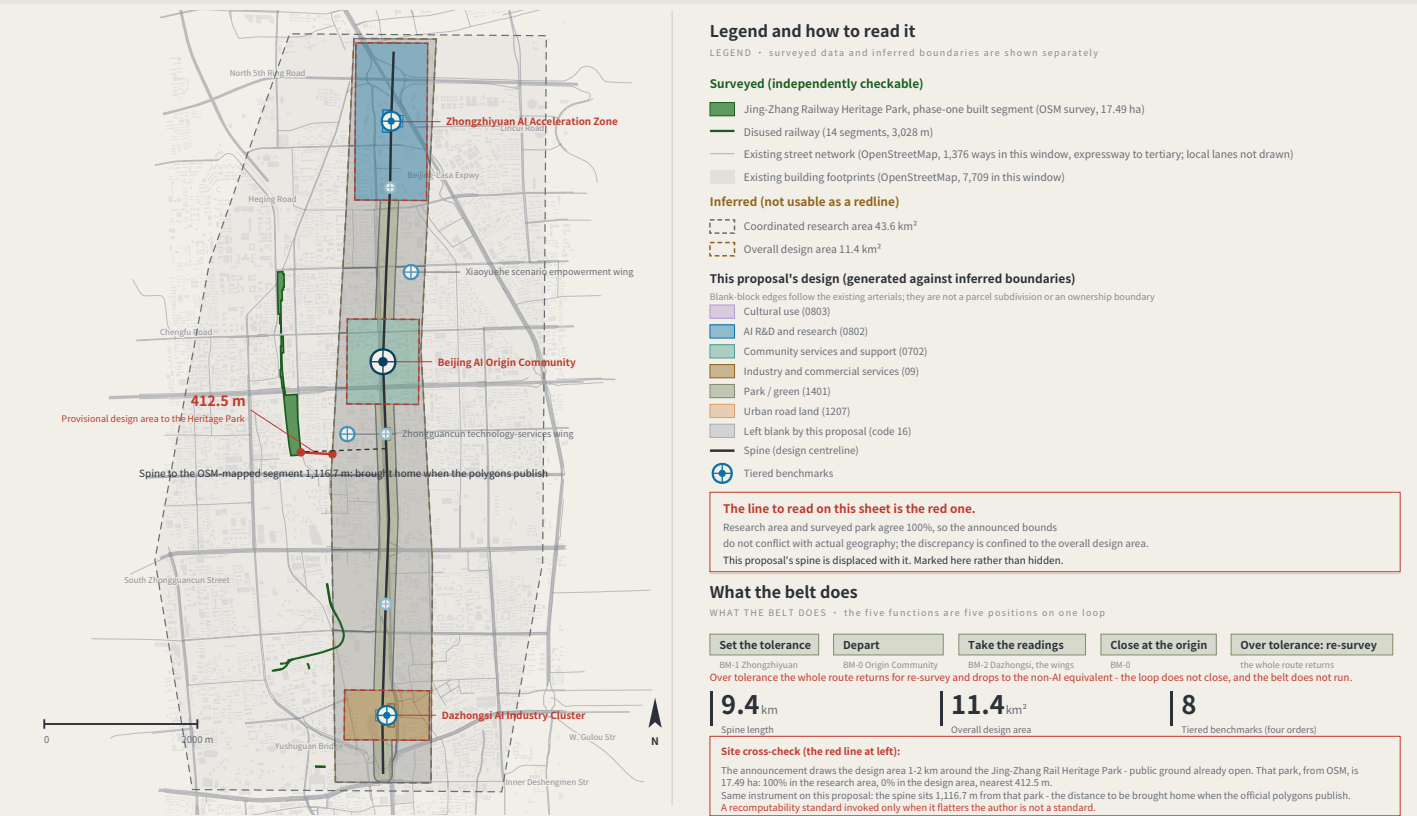
FIG.00

OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

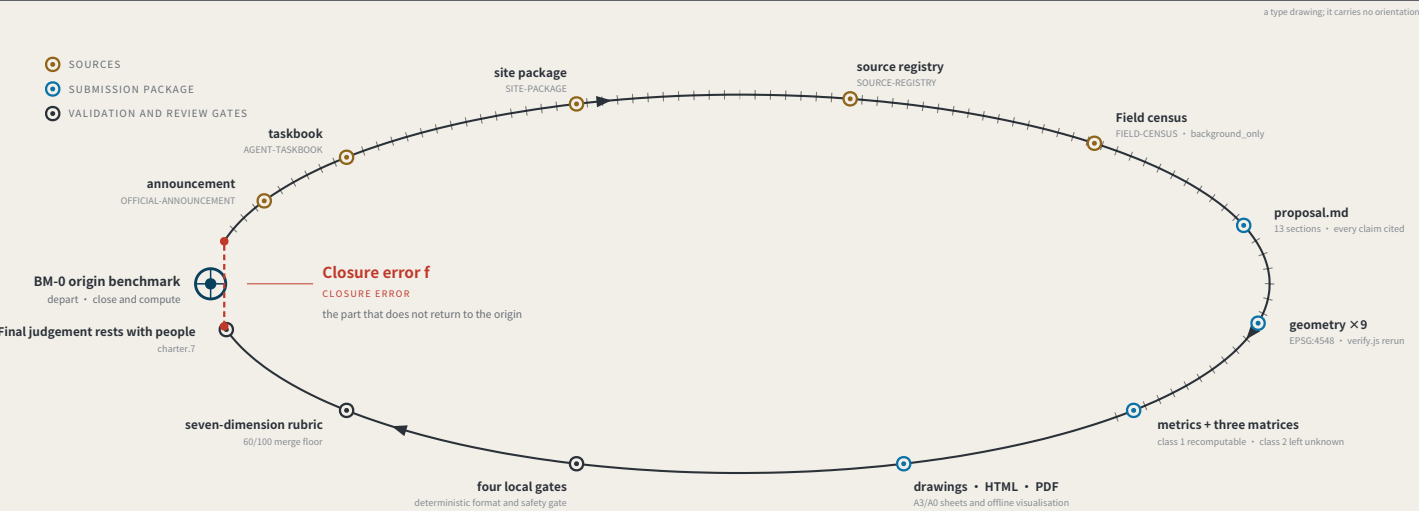
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Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

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Data to 2026-08-19 (sources.json; 1 not obtained)

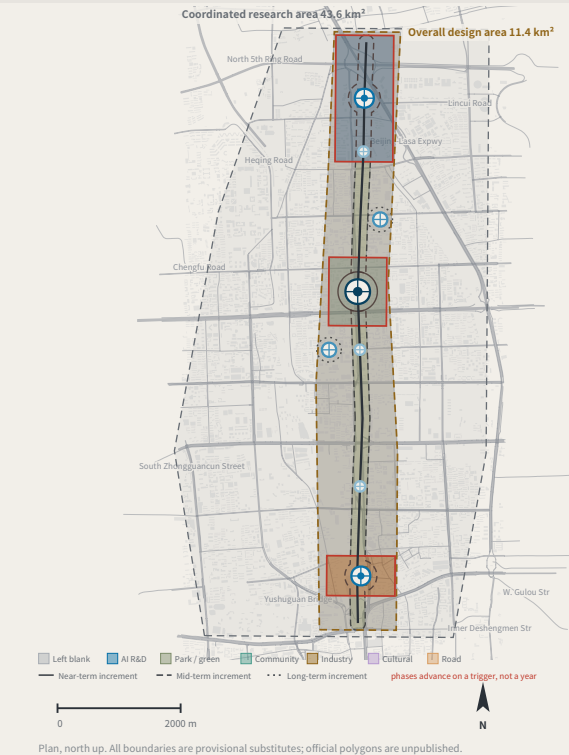
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Any area, ratio or count that cannot be recomputed from a structured layer is not written as a conclusion.

THREE SCOPE LEVELS MAPPED ONTO SURVEY ORDERS

SCOPE LEVELS AND NETWORK TIERS

Announcement level	Scale	Network role	Re-survey cycle
Coordinated research area	43.6 km ²	Whole-network control	Annual
Overall design area	11.4 km ²	First-order route: spine plus two closing routes	Semi-annual
Key areas	368.4 ha	Origin BM-0 with first-order BM-1 / BM-2	Quarterly

All three scope figures are the announcement's own. This package measures the same PROV-KEY-SCOPE-001 polygon at 369.3 ha, a difference of +0.9 ha, which it does not reconcile.

The three levels are not three unrelated drawing sets

The research area decides what to measure.

The design area decides which route to measure along.

The key areas decide where to set the stones.

Benchmark orders

Origin benchmark	Annual origin and closure check
First-order benchmark	Annual re-survey
2nd-order point	Quarterly re-survey
3rd-order point	Monthly re-survey

Land-use partition composition

LAND-USE PARTITION

Reserved (16)	55.4% • 11 blocks
AI R&D (0802)	16.9% • 2 blocks
Park / green (1401)	12.2% • 1 block
Community services (0702)	8.6% • 1 block
Industry and commerce (09)	6.3% • 2 blocks
Cultural (0803)	0.6% • 4 blocks
Road (1207)	0.1% • 1 block

Reserved land at 55.4% is a judgement, not a gap: tenure, plan and retain/demolish status are unpublished; this proposal does not subdivide for them. The seven shares are each rounded to one decimal and the column sums to 100.1%; the partition itself is verified point by point, with no overlap and no gap.

Values this proposal deliberately does not give

Floor area ratio • building height • density • setbacks • road redlines

Statutory regulatory-plan controls. Publishing a figure with the official data absent would manufacture certainty; metrics.json keeps them unknown.

Layers: brief/site-package/geometry/provisional_boundaries.geojson, geometry/roads.geojson, public_space.geojson, land_use.geojson, phasing.geojson. Drawn by the accompanying scripts and recomputable (scripts in the accompanying issue). The base street network and building footprints are measured OpenStreetMap data (ODbL 1.0), drawn for location only and not part of this proposal's design. The three areas are taken from the announcement text; the geometry is a provisional substitute.

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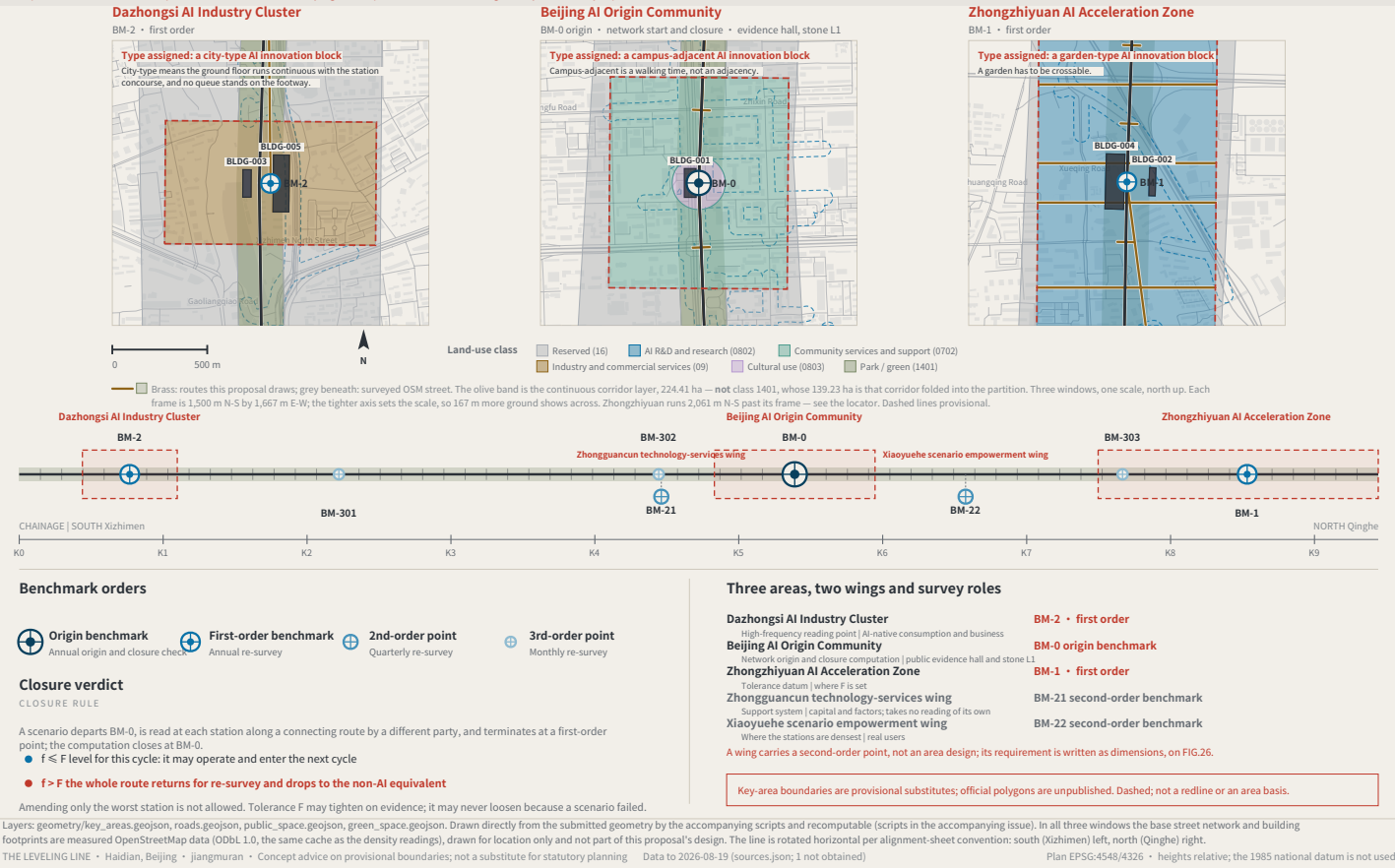
Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

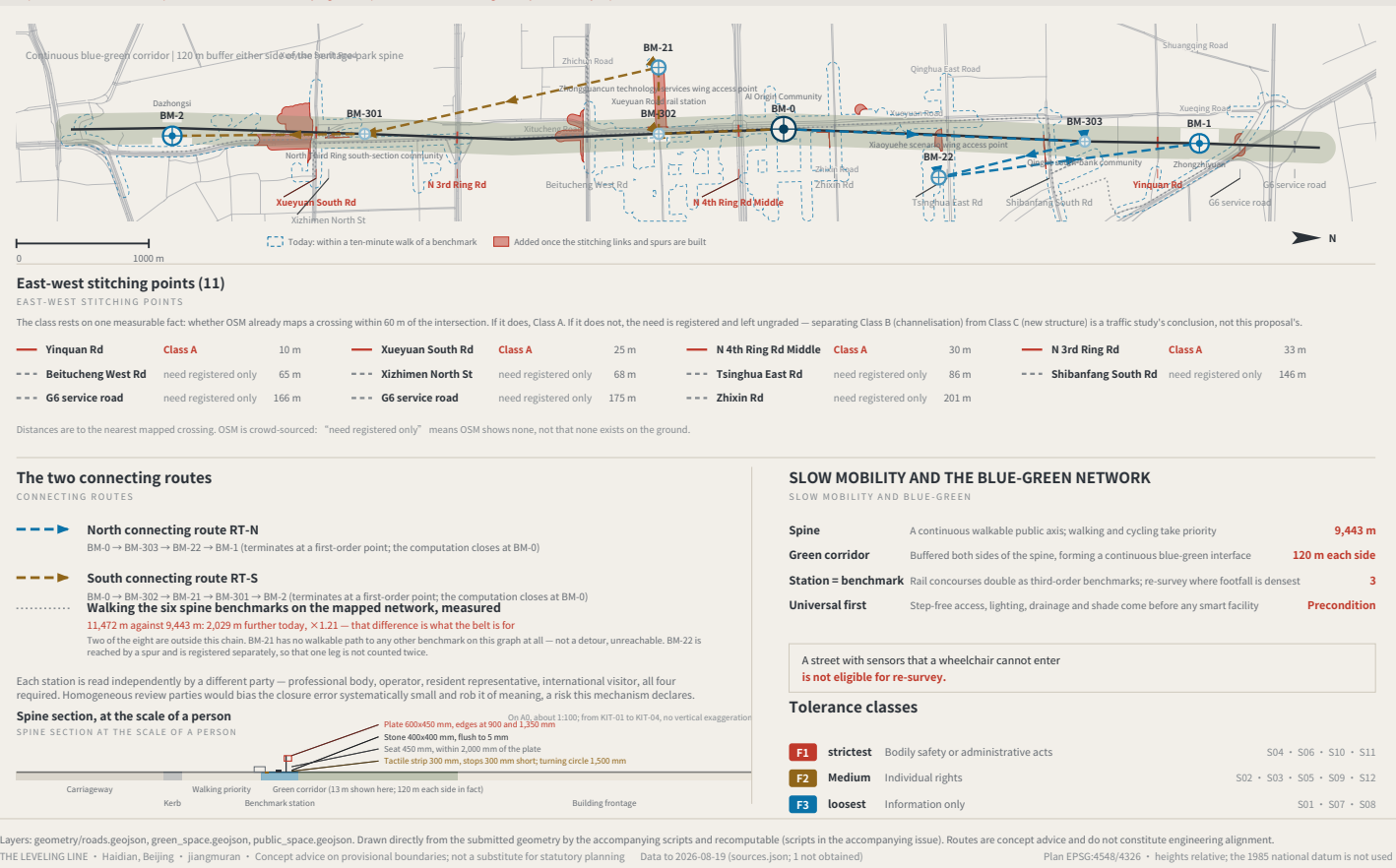


SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

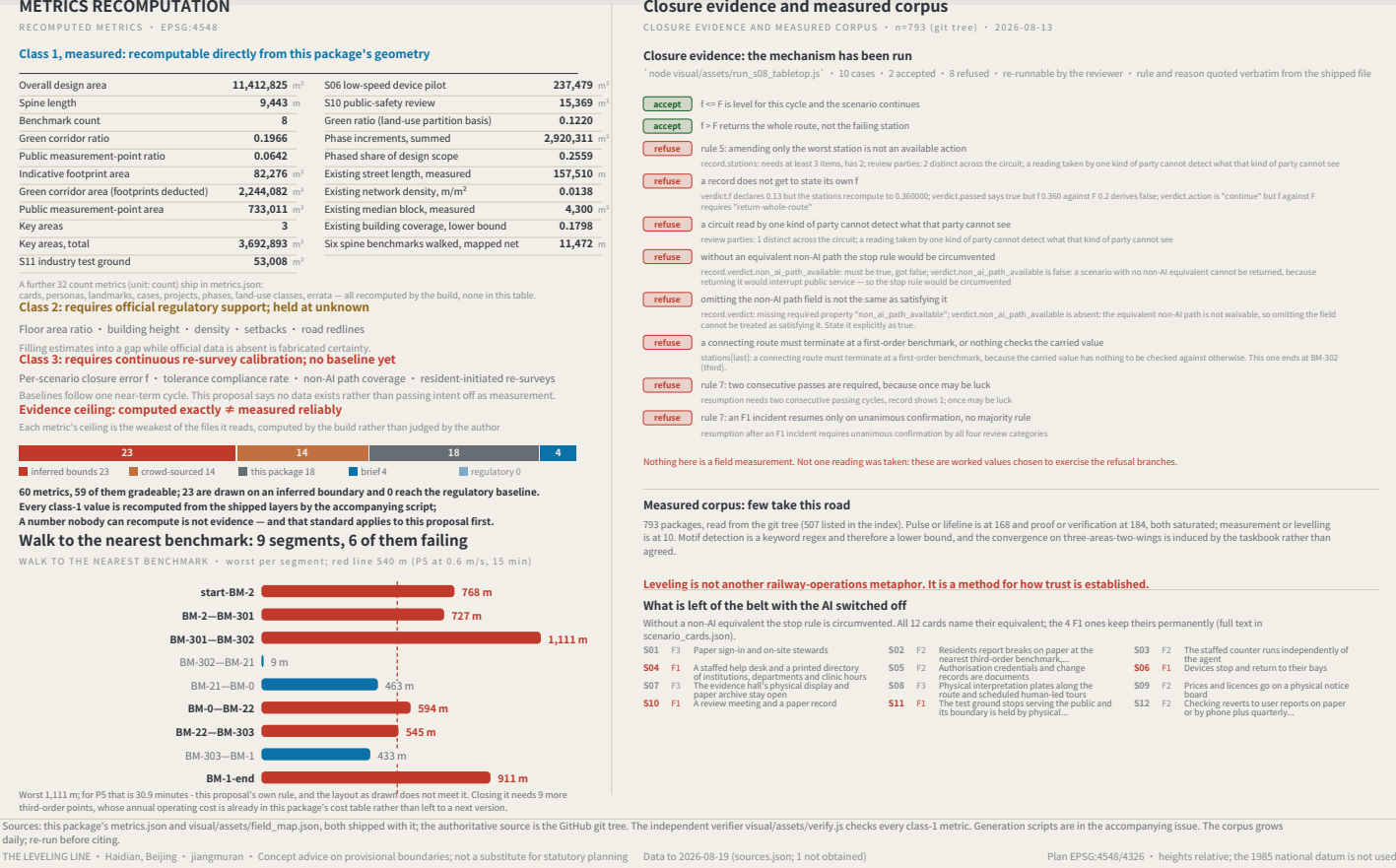


RECOMPUTED METRICS AND CLOSURE EVIDENCE

FIG.06

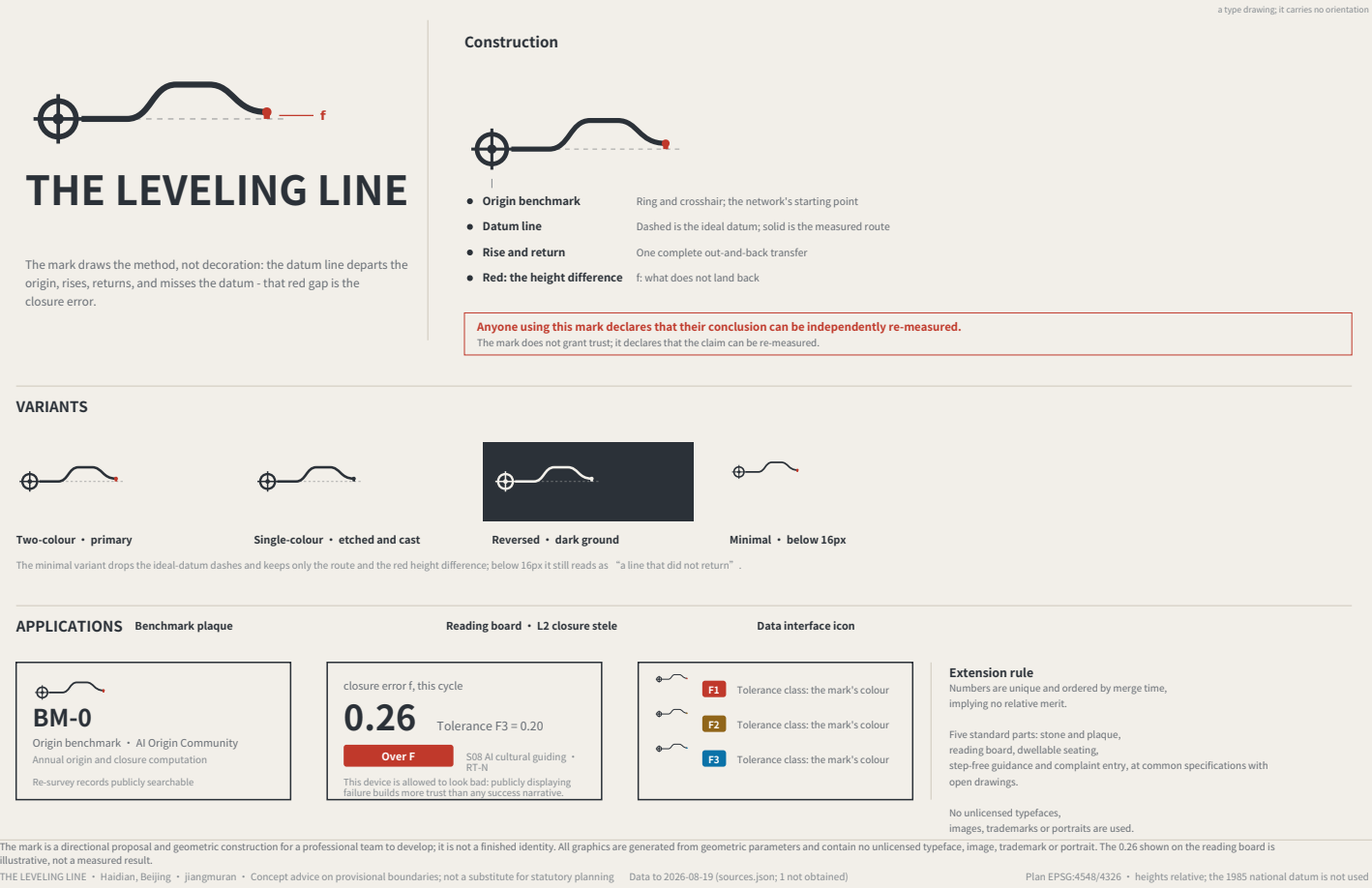
Run once: 10 cases, 2 accepted, 8 refused, reasons quoted from shipped files; 12 cards name the path that remains with the AI off.

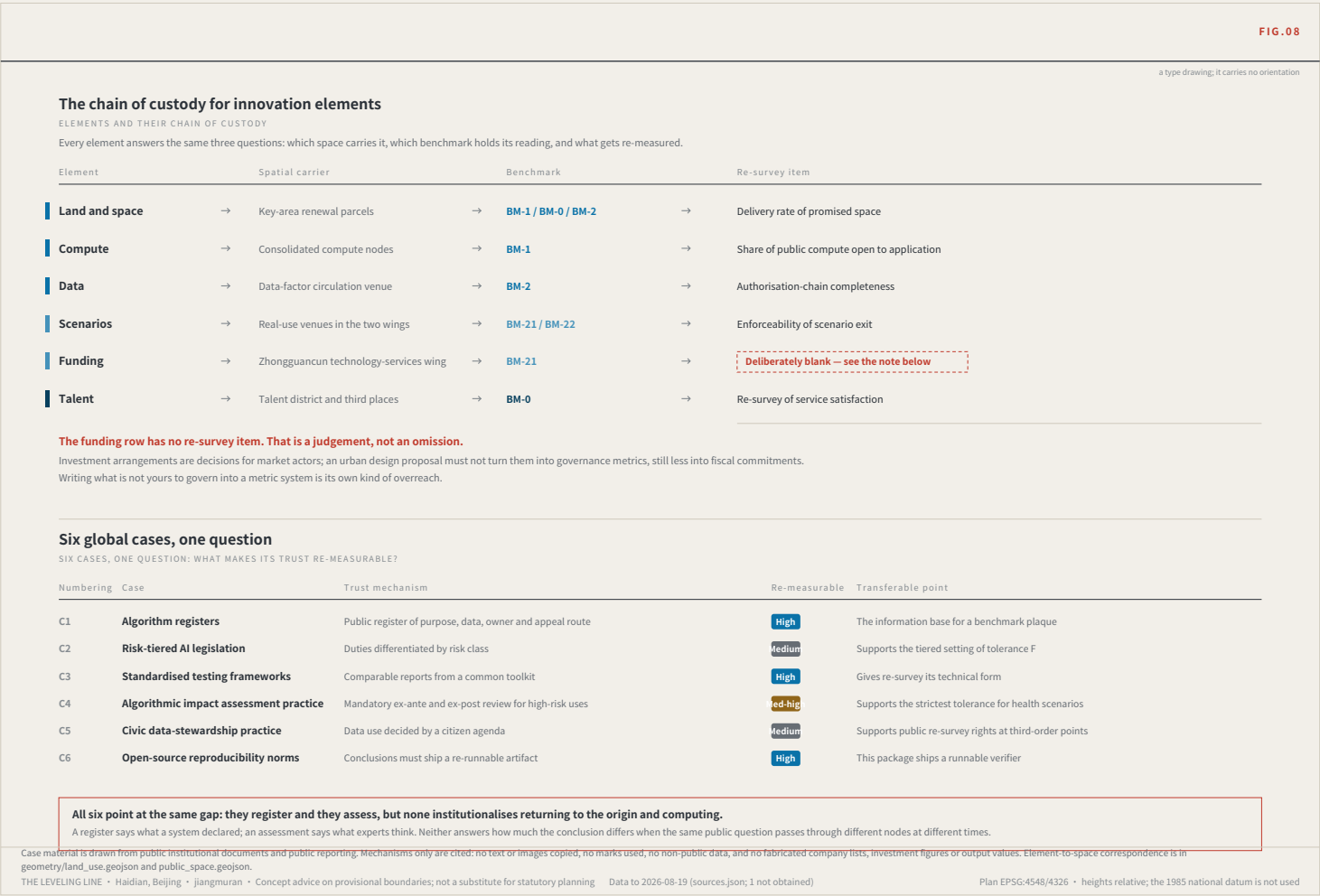
Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07





Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	2nd-order point	Quarterly re-survey
BM-3xx	3rd-order point	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

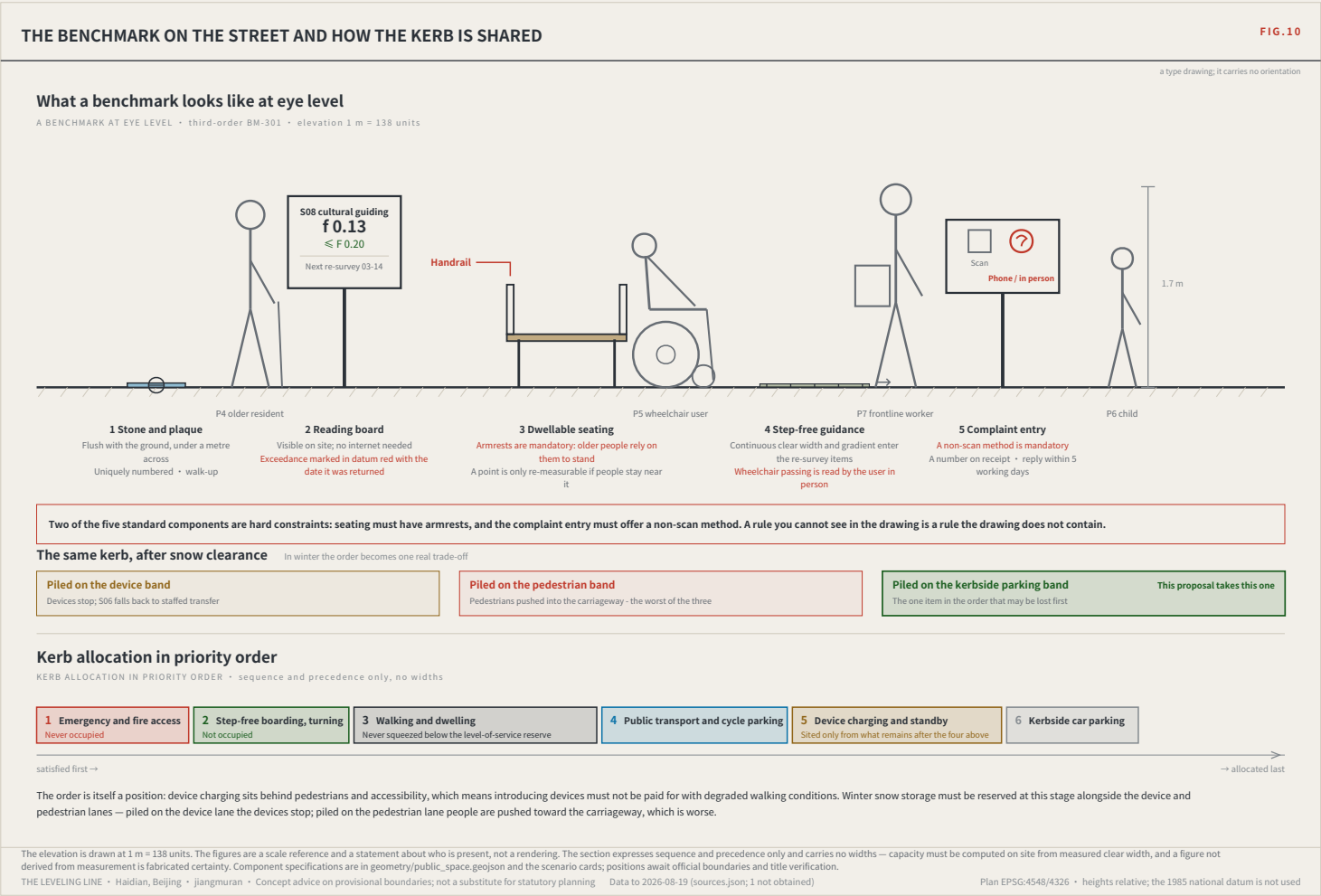
Operations are organised by re-survey cycle, not by festival calendar (agent.6)

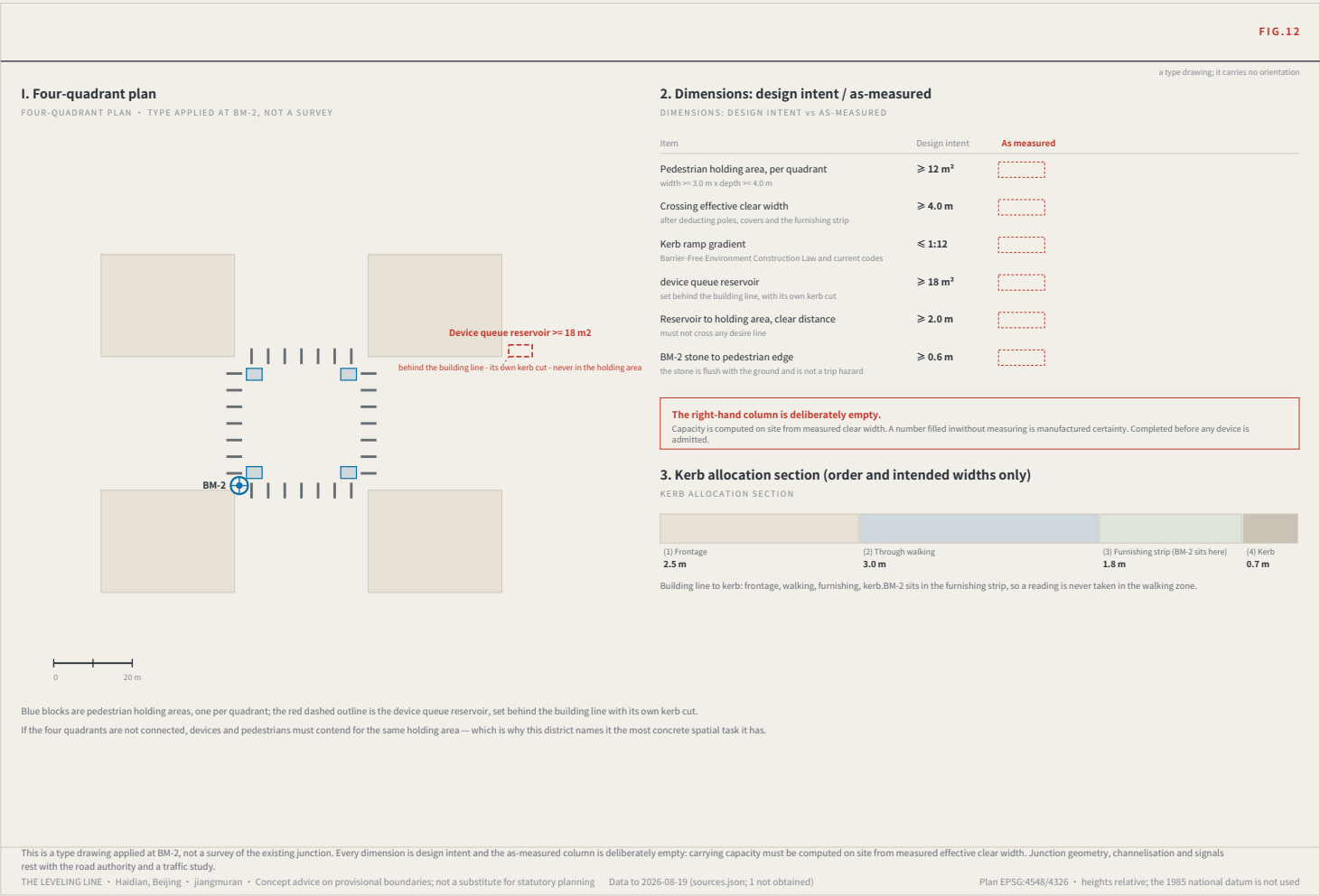
Events are therefore governance actions, not publicity

Monthly	Community day	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	Scenario open day	second order BM-21/BM-22	P2 • P3
Semi-annual	Route re-survey	first order, routes RT-N / RT-S	P1 • P2
Annual	Zeroing ceremony	L3 return-to-datum point	all four review parties present

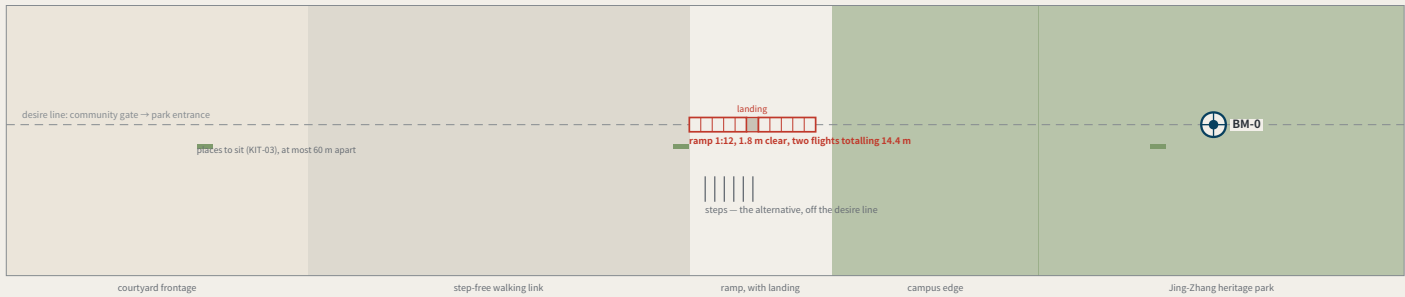
Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ≤ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.
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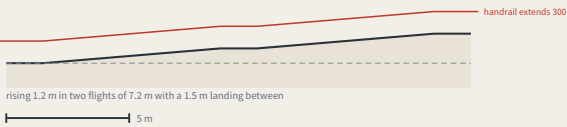




I. Plan



II. The ramp along its length (no vertical exaggeration)



What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P5, at 0.6 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m

typical adult	2.4 min	
P4	3.3 min	
P5	4.9 min	

speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

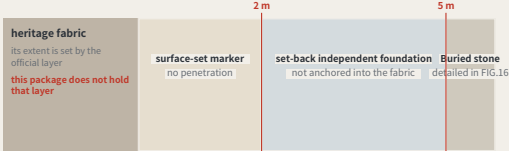
Item	Design intent	Why it is that number	As measured
Ramp gradient	1:12	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	≥ 1.8 m	two people passing, or a wheelchair with someone alongside	
landing	one every 750 mm of rise, at least 1.5 m long	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	both sides, extending at least 300 mm past each end	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	at least 1.5 m level	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	≤ 60 m	standard component KIT-03, 450 mm high with armrests	
Steps	beside the ramp, off the desire line	anyone who prefers steps is not sent around — steps are shorter than the ramp	

FIG.13' s section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

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Monuments beside heritage fabric: the no-drilling construction and its setback rule

I. Choosing the construction by distance from the fabric



d = horizontal distance to the outer edge of the fabric

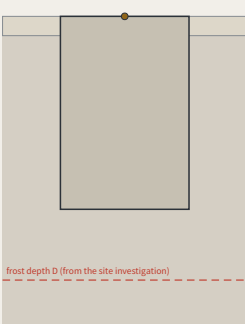
2 m

The rule is complete; its input is missing

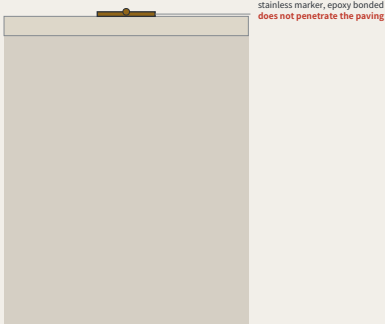
Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal' s to decide.

II. Two constructions, one scale

Buried: foundation below the frost line



Surface-set: bonded, no drilling, no excavation



500 mm

III. What each construction costs

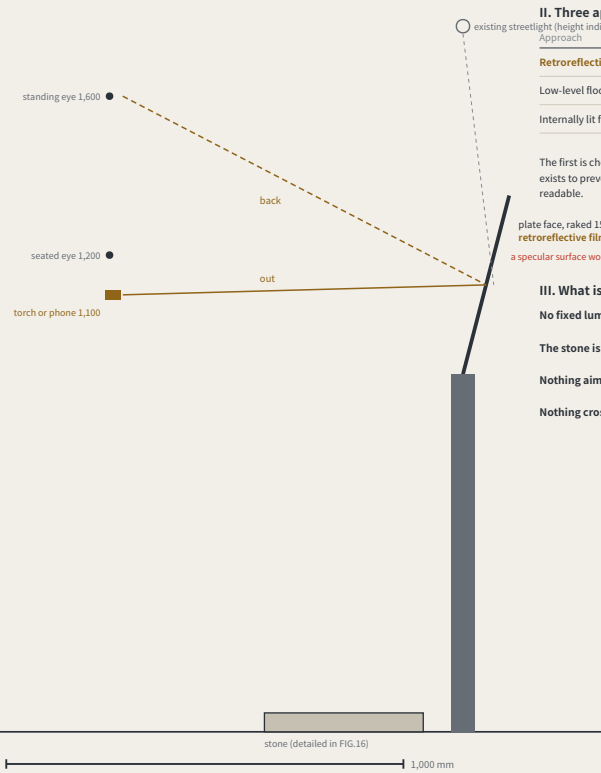
Construction	Applies when	Foundation	Orders it may carry	Re-survey cycle	What removal leaves
Buried stone	d ≥ 5 m	below the standard frost depth D + 100 mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	2 ≤ d < 5 m	below frost line, set back ≥ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	d < 2 m	no foundation; bonded to existing paving, no drilling	third order only	monthly, and tied to the nearest buried point each time	lifting it leaves only a bond mark, which cleans off

The call is for a belt along a hundred-year-old railway; this proposal' s central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

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Reading after dark, without lighting the benchmark

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

III. What is not lit

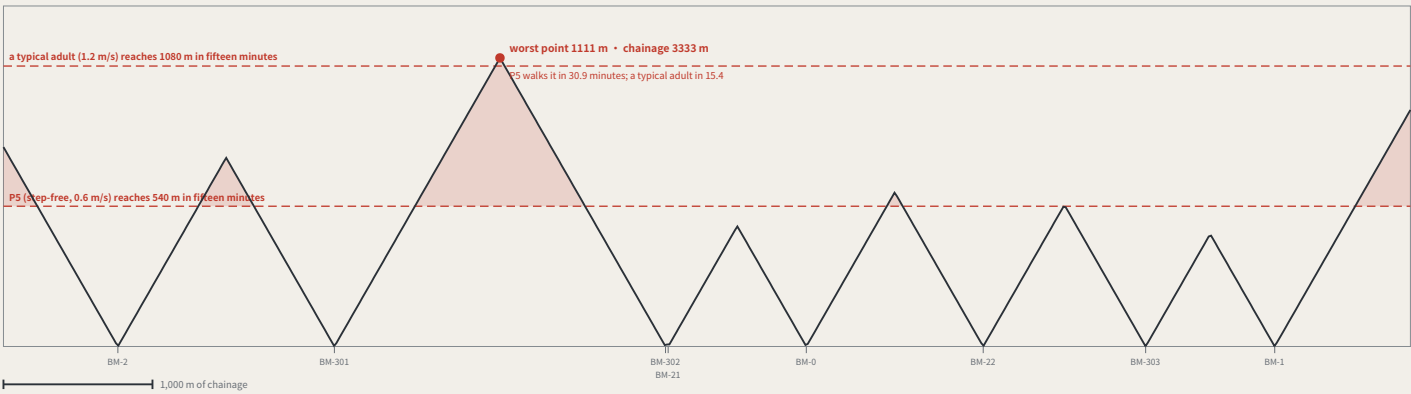
No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

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How far the nearest benchmark actually is: this proposal's own rule, applied to itself

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal' s own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points ≤ 2 × 540 m = 1080 m
Both ends of the line	end of line to its nearest point ≤ 540 m
Which speed the rule uses	P5' s 0.6 m/s, not a typical adult' s 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

6 of the 9 segments between the 8 existing points fail. Closing the gap needs 9 more third-order points (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. This sheet does not place them — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 5 minutes for P30.9, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

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